Assignment 1 - Phase 1 - Saurav Dasgupta C52

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Background

As a part of Assignment 1 on **Product Sense**, provided by HelloPM, we were provided with the below Problem Statement:

Pick one product you use, investigate it properly, learn from the people who use it, find the one problem worth solving, and design the answer.

This assignment is divided into 3 phases:

Phase	Objective	Description
Phase 1	**Understand the Product and Users**	Study the product with fresh eyes, work out how the business behind it earns, map the different kinds of people who use it, and list every problem you can see. Ends with a goal, and 1 to 3 problems to take into research.
Phase 2	**Research & Learn**	Take your problems to real people. Run interviews and a survey, find out how users actually behave rather

Phase	Objective	Description
		than how you assumed they would, and stay open to what you did not expect. Ends with one sharp P0 problem, backed by evidence.
Phase 3	**Solve & Communicate**	Brainstorm widely against your P0, prioritize down to what is worth building, prototype it, decide how you would measure success, and plan for what could go wrong. Ends with a working prototype and a complete PRD.

This document deals with Phase 1 of the assignment for the product I have chosen.

My Product of Choice

For this assignment, I have picked **YouTube** as my product of choice.

Research and Analysis of the Product and the Business

Observing the Product as a real viewer

What I Tried	What Actually Worked	Where I had to Work
Using the Product as a Guest User without signing in	• Video Search • Video Playback • Basic content recommendation based on the current session	• Was not provided recommended content based on previous usage after reinstalling the app • Couldn't revisit watch history, in order to watch previously viewed content • Was not able to Like/Dislike videos • Was not able to comment on videos • Was not able to save videos offline • Was not able to subscribe to channels of preference
Using the Product as a	• Was provided with	Was not provided with Recommended Content from

What I Tried	What Actually Worked	Where I had to Work
Logged In user after signing into my Google Account	Recommended Content from previous logged in sessions • Could revisit Watch History, when I felt like watching previously viewed content from previous logged in sessions • Was able to Like/Dislike videos • Was able to comment on videos • Was able to save videos for offline viewing.	my previous usage of the app as a Guest User.
Using the Product on a secondary device with a different Google account.	I was able to enjoy all the features of a logged in user	As a user, I often felt limited trying to look up some video that I had viewed on my primary device with my primary account logged in. I wish I could have continued my user experience of using Youtube on my primary device on my secondary device as well. I had to manually search for content of my 'Liked' creators (that I had liked on my Primary account) and had to re-do offline downloads of my favourite content from my primary account. My only option was to login with my primary account on my secondary device as well and to keep switching between accounts on Youtube. However, since my secondary device is used only for my side-business, and I prefer to keep my personal and work lives separate, that was not an option for me.

Reading the Business

Question	What I found
How it makes money	Advertisements, YouTube Premium subscriptions, Channel memberships, Shopping and affiliate features, Movie purchases and rentals
Who pays, and for what	Advertisers pay the platform for advertising their business, also viewers/consumers pay YouTube for YouTube Premium subscriptions, channel memberships, shopping and affiliate features , movie purchases and rentals
Recent focus	1. AI powered creation tools 2. Greater monetization for creators 3. TV and living room experience 4. AI backed recommendations 5. Community and fan engagement 6. Content Integrity

Competitor Analysis

Competitor Read

Competitor	Category	Does Better	Does Worse
TikTok	Direct Competitor	Better at short form content such as Reels. More addictive than YouTube with “For You” feed that surfaces engaging content quickly. Has better, easier-to-use editing features which make content creation relatively simpler compared to YouTube.	Search feature is way worse than YouTube. Monetization streams limited compared to YouTube. Limited long form content
Netflix	Direct Competitor	Better content for entertainment, richer recommendation system.	Being a subscription based model, the content can be

Competitor	Category	Does Better	Does Worse
		More collection of movies and shows contribute to a richer viewer experience. Original shows which cannot be found on other free platforms like YouTube due to copyright protection	accessed only after paying a subscription fee. There is no user generated content on Netflix, as it is not an open platform where anyone can upload videos unlike YouTube. No virtual interaction with creators through comments, emojis or stickers - there is no community as such
Udemy	Indirect Competitor	Better structured content, reviewed by peers, bigger community and interaction with the educators through forums and Q&As. Keeps viewers motivated and tracks their progress by maintaining steaks, giving reminders and also helps them to assess their learning by quizzes built into their content	Only educational or professional content - rather single dimensional. Most of their highly rated content has to be purchased or need to be accessed through a subscription fee

Where people go instead and when

The Moment	Where They Go	What They Get There
To watch addictive, viral short form content	Instagram Reels and TikTok	More addictive and fresh content with superior content discovery and a better recommendation system for the viewers
To upload short form content with the intention	TikTok	Easier to get viral and famous as compared to YouTube, since it is

The Moment	Where They Go	What They Get There
to go viral and famous		heavily organized around what users like , rather thanwho they follow
To seriously learn and consume educational content for academic and professional reasons	Udemy and Coursera	Better structured learning content with learning paths, with periodic in-content quizzes to assess progress, better vetted courses due to ratings and reviews . More probability of actually sticking through and completing it due to timely reminders and streaks

What is changing in the category

- AI-assisted creation is becoming central to the product (auto-editing, captions, translation, idea generation).
- Short-form remains a primary discovery surface, even for creators who monetize via long-form.
- Consumption is shifting to the living room (smart TVs, game consoles).
- Creator monetization is diversifying (subscriptions, memberships, shopping/affiliates, tips).
- Trust & safety + content integrity are under tighter scrutiny (misinformation, deepfakes, copyright).
- Communities are moving from passive comments to richer interactions (live chat, communities, creator–fan tools).
- Recommendation quality is an important battleground for all products in this category.

User Segments

Primary Segment	Sub-Segment	What They Need from the Product
Content Creators	Long Video Creators	Easy video upload, reliable monetization, audience growth, discoverability, analytics, copyright protection, editing tools, audience engagement (comments, memberships, live streams), collaboration tools, AI-assisted content creation.
Content Creators	Independent Artists and Production Houses	High-quality video hosting, content rights management, content distribution, brand visibility, monetization through ads/premium, licensing support, promotional tools, audience analytics.
Content Creators	Shorts Creators	Fast content creation, mobile-first editing, viral discovery, trending music/effects, rapid publishing, creator insights, easy monetization, AI editing tools.
Content Creators	Independent musicians and Studio labels	Music distribution, royalty management, copyright protection (Content ID), fan engagement, music discovery, playlist visibility, revenue from ads and subscriptions, promotional campaigns.
Content Creators	Podcast Creators	Audio & video podcast hosting, RSS integration, monetization, audience analytics, discoverability, chapter support, transcription, live streaming, community engagement.
Content Creators	Educators	Structured learning content, playlists/courses, quizzes (if available), reliable search, audience retention insights, monetization, captioning, multilingual support, classroom/community interaction.
Content Viewer	Long format Video Viewer	Personalized recommendations, high-quality streaming, watch history, playlists, subtitles, offline viewing, minimal buffering, cross-device sync, relevant search, meaningful recommendations.

Primary Segment	Sub-Segment	What They Need from the Product
Content Viewer	Shorts Viewer	Endless personalized feed, quick entertainment, fast loading, intuitive swiping, relevant recommendations, easy sharing, interactive features (likes, comments, remix).
Content Viewer	Music Listeners	High-quality music streaming, playlists, recommendations, background play, offline downloads, lyrics, personalized mixes, artist discovery.
Content Viewer	Podcast Viewers	Easy podcast discovery, background playback, chapter navigation, transcripts, offline listening, personalized recommendations, playback speed controls.
Content Viewer	Students and Professionals	Accurate search, trustworthy educational content, bookmarks, playlists, note-taking support, transcripts, subtitles, playback speed control, AI summaries, recommendations based on learning goals.
Advertisers	Small Businesses	Affordable advertising, simple campaign creation, audience targeting, measurable ROI, budget control, conversion tracking, AI campaign recommendations.
Advertisers	Large Enterprises	Advanced audience targeting, large-scale campaign management, brand safety, cross-channel analytics, attribution models, API integrations, performance optimization, dedicated support.
Advertisers	Media Agencies	Multi-client campaign management, reporting dashboards, audience insights, campaign automation, bulk operations, collaboration features, optimization tools, API access, transparent performance metrics.

Problems That I Have Hypothesized a User Would Face

The following problems are hypotheses derived from observing YouTube's product experience, studying its ecosystem, comparing it with competitors, and conducting secondary research. These hypotheses will require validation through user interviews and usability research before they can be considered genuine user pain points.

Problem (as a user would say it)	Who feels it? (User Segment)	Points to
1. I spend too much time searching before I find a video that actually answers my question.	Students, Professionals, Knowledge seekers	**Engagement, Retention, Search Quality**
2. YouTube keeps recommending the same type of videos even after my interests have changed.	Frequent viewers	**Engagement, Personalization, Retention**
3. I often skip through long videos trying to find one specific answer. I wish YouTube could summarize or let me search inside videos.	Students, Professionals	**Engagement, Watch Time, Satisfaction**
4. I don't always trust the videos YouTube recommends because some contain misleading or factually incorrect information.	News consumers, Educational viewers	**Trust, Retention, Brand Reputation**
5. I wish YouTube clearly indicated when a video may contain disputed or potentially misleading information.	All viewers	**Trust, Safety, User Confidence**
6. Sometimes subtitles aren't available in my language,	International viewers	**Accessibility, Engagement, International Growth**

Problem (as a user would say it)	Who feels it? (User Segment)	Points to
preventing me from watching otherwise useful content.		
7. Auto-generated subtitles and translations are often inaccurate, especially for technical videos and regional accents.	Students, Professionals, Hearing-impaired users	**Accessibility, Retention, Educational Experience**
8. Different members of my family use the same account, which completely messes up my recommendations.	Families, Shared-device households	**Personalization, Engagement, Retention**
9. I lose my personalized recommendations and watch history when I use YouTube as a guest or switch devices.	Guest users, Multi-device viewers	**Retention, User Activation, Cross-device Experience**
10. YouTube Kids doesn't adapt well when children of different ages share the same device.	Parents with multiple children	**Safety, Personalization, Family Retention**
11. As a creator, I don't understand why one of my videos performs well while another doesn't.	Content creators	**Creator Retention, Content Supply, Satisfaction**
12. As a creator, I wish YouTube provided better AI tools for editing, thumbnails, titles and content ideas.	Content creators	**Creator Productivity, Content Supply, Engagement**
13. As a creator, YouTube's moderation and monetization decisions sometimes feel unpredictable and poorly explained.	Monetized creators	**Creator Trust, Creator Retention, Revenue**

Problem (as a user would say it)	Who feels it? (User Segment)	Points to
14. I don't feel YouTube Premium offers enough value beyond removing advertisements.	Premium subscribers	**Revenue, Subscriber Retention, Customer Satisfaction**

Identifying the Problem Clusters

By analyzing the affected user segments and the business metrics associated with each problem, several clear product opportunity areas emerge.

Problem Cluster	Representative Problems	Primary Business Metrics
Content Discovery & Search	Problems 1, 2, 3	Engagement, Watch Time, Retention
Trust, Safety & Content Integrity	Problems 4, 5, 13	Trust, Retention, Brand Reputation
Accessibility & Global Reach	Problems 6, 7	Engagement, Accessibility, International Growth
Household & Multi-user Experience	Problems 8, 9, 10	Retention, Personalization, Family Experience
Creator Experience & Growth	Problems 11, 12, 13	Creator Retention, Content Supply
Subscription & Monetization	Problem 14	Revenue, Subscriber Retention

Where Do the Problems Cluster?

Although the hypotheses span multiple areas of the YouTube ecosystem, the strongest concentration lies within **Content Discovery & Search**, with several additional problems closely related to **Recommendation Personalization**.

A common pattern emerges:

- Users struggle to **find the right content** efficiently.

- Users struggle to **receive recommendations that reflect their current interests**.
- Users struggle to **extract useful information from long-form videos quickly**.

These challenges all contribute to the same underlying business outcomes:

- Lower engagement
- Reduced watch time
- Lower session duration
- Decreased viewer retention
- Reduced long-term satisfaction

Product Opportunity Category

Content Discovery & Recommendation Personalization

Proposed Product Goal

> **Improve engagement and retention for knowledge-seeking viewers by helping them discover relevant, trustworthy content faster and with less effort.**

This goal aligns with YouTube's core value proposition while supporting several of its most important business metrics, including watch time, session duration, viewer retention and recommendation quality.

Problems to Probe

To validate the proposed product goal, I would prioritize the following three hypotheses for user research.

Problem 1

> **"I spend too much time searching before I find a video that actually answers my question."**

Why this problem?

This problem affects a broad audience of students, professionals and everyday learners. If validated, solving it could significantly improve search efficiency, engagement and overall user satisfaction.

Problem 2

> **"YouTube keeps recommending the same type of videos even after my interests have changed."**

Why this problem?

Recommendations are central to YouTube's user experience. Understanding whether users feel trapped in repetitive recommendation loops could uncover opportunities to improve personalization and long-term retention.

Problem 3

> **"I often skip through long videos trying to find one specific answer. I wish YouTube could summarize or let me search inside videos."**

Why this problem?

As long-form educational and informational content continues to grow, users increasingly value speed and efficiency. Validating this hypothesis could reveal opportunities for AI-powered summaries, semantic search and improved in-video navigation, leading to higher engagement and greater consumption of long-form content.

Research Objective

The primary objective of the research is to determine whether users experience significant friction in discovering and consuming relevant content, understand how these issues affect engagement and retention, and identify opportunities where YouTube can leverage AI and personalization to deliver a faster, more relevant and more satisfying viewing experience.